Abby Joseph

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EDUCATION

Graduation: May 2027

MSE in Robotics (Perception and Cognition Track)
Johns Hopkins University

August 2021-May 2025

BS in Computer Engineering (Electronic Systems Track)
University of Maryland, Baltimore County
GPA: 3.926

INTERNSHIPS AND RESEARCH EXPERIENCE

January 2026 – Present

Agile and Intelligent Robotics Laboratory
Johns Hopkins University
Principal Investigator – Joseph Moore

- Used reinforcement learning (RL) to dynamically follow a goal for future RL-based formation control in an obstacle filled environment

June 2025 - Present

Embedded Electronics Intern – GEM Fellow
Draper Laboratory

- Collaborated to design a PCB for power distribution
- Conducted JTAG testing to validate interconnect for custom electronics
- Used HDL to design digital systems synthesized for custom electronics testing

June 2023-May 2025

Interactive Robotics and Language Lab
University of Maryland, Baltimore County
Principal Investigator – Cynthia Matuszek Ph.D.

- Helped develop coordinated manipulation of two arms for a human-robot collaborative task of the assembly of a three-piece toy (in VR and reality).
- Investigated algorithms to dynamically determine the next-best view for a camera platform observing a human-robot collaborative assembly task

June 2024 - August 2024

Laboratory for Computational Sensing and Robotics
Johns Hopkins University
VTSI
Principal Investigator – Ralph Etienne-Cummings Ph.D.

- Used machine learning with handcrafted features to correlate bio-signals for progress towards non-invasive continuous blood pressure monitoring

HONORS AND AWARDS

- GEM Employer Fellowship with Draper Laboratory and Johns Hopkins University
- UMBC Meyerhoff Scholar

TECHNICAL INVENTORY

Languages: Python, C/C++, MATLAB, SystemVerilog, VHDL, BSDL

Software: Robot Operating System (ROS), Xilinx, PSpice, Cadence Virtuoso, Git/Github, Linux, Mission Planner

Skills: Arduino Microcontrollers, Field Programmable Gate Arrays (FPGAs), VLSI, MAVLink

EXTRA-CURRICULARS

Summer 2026 – Spring 2027: Whiting School of Engineering – Master's Peer Navigator

Fall 2021-Spring 2025: UMBC Varsity Women's Soccer

MEMBERSHIPS

- Tau Beta Pi (Engineering Honor Society)
- Chi Alpha Sigma (National College Athlete Honor Society)